

Notes: Cunningham, Ederer, and Ma (JPE, 2021)

Killer Acquisitions

Contents

1	What question does the paper answer?	3
1.1	Main question	3
1.2	Operational definition: “killer acquisition”	3
2	Theoretical framework (all model objects and results)	3
2.1	Timeline (Figure 1)	3
2.2	Product-market profits and overlap	3
2.3	Development decision at $t = 1$ and the replacement effect	3
2.4	Threshold form of development choices (eqs. (3)–(4))	4
2.5	Proposition 1: overlap reduces post-acquisition development	4
2.6	Proposition 2: competition mitigates the replacement effect	4
2.7	Proposition 3: longer patent life strengthens killer incentives	4
2.8	Acquisition decision at $t = 0$: efficiency effect vs replacement effect (eqs. (5)–(6))	5
2.9	Proposition 4: overlap predicts acquisition	5
2.10	Multiple incumbents (sketch)	5
3	Data and measurement (how the paper maps theory to empirics)	5
3.1	Setting: pharmaceutical R&D	5
3.2	Project data: Pharmaprojects	6
3.3	Acquisition data and merging	6
3.4	Overlap measure (core identification variable)	6
3.5	Competition environment	6
3.6	Patent life	6
3.7	Antitrust scrutiny: HSR threshold	6
4	Empirical strategy: all econometric models	7
4.1	Model (7): project-year development panel (main test)	7
4.2	Competition heterogeneity (Table 4)	7
4.3	Patent-life heterogeneity for overlapping acquisitions (Table 5)	7
4.4	Model (8): acquisition choice using conditional logit (Table 6)	7
4.5	Antitrust threshold discontinuity evidence (Table 7 and Figure 3)	8
4.6	Asset redeployment and human-capital channel (Table 8)	8

4.7	Acquisition value regression (Table 9)	8
5	Results	8
5.1	Figure 1: model time line	8
5.2	Figure 2: optimal acquisition strategies	9
5.3	Table 1: descriptive acquisition rates	9
5.4	Table 2: main development result (equation (7))	9
5.5	Table 3: robustness and “never developed” evidence	10
5.6	Table 4: competition heterogeneity (Proposition 2)	10
5.7	Table 5: patent-life heterogeneity (Proposition 3)	11
5.8	Table 6: overlap predicts acquisition (Proposition 4)	11
5.9	Figure 3 and Table 7: antitrust threshold evidence	11
5.10	Table 8: post-acquisition asset redeployment and human capital	11
5.11	Table 9: overlap and acquisition value	12
5.12	Back-of-the-envelope prevalence of killer acquisitions (from text around Table 9)	12
6	What to remember (exam-style takeaways)	13
7	Conclusion	13
7.1	Summary	13
7.2	Key empirical conclusions	13
7.3	Quantitative takeaway: how prevalent are killer acquisitions?	14
7.4	Why overlap creates “acquire-to-kill” incentives	14
7.5	Why this matters for antitrust and IO inference	15
7.6	Takeaway	15

1 What question does the paper answer?

1.1 Main question

When an incumbent acquires an innovator (often early in R&D), does the acquisition tend to *accelerate* innovation (synergies, better development capabilities) or can it instead *reduce* innovation by **shutting down** the target's project to preempt competition?

1.2 Operational definition: “killer acquisition”

An acquisition is a *killer acquisition* if it is motivated by preempting future competition and is followed by discontinuation/termination of the acquired innovation project. Because direct termination reporting is incomplete, the paper measures termination using **lack of subsequent development events** in a project-year panel.

2 Theoretical framework (all model objects and results)

2.1 Timeline (Figure 1)

- $t = 0$ (acquisition stage): an entrepreneur E owns a single project; $n \geq 1$ incumbents exist; one incumbent A can acquire E at price P .
- $t = 1$ (development decision): the project owner chooses whether to continue development. Development succeeds with probability r_A if A develops or r_E if E develops; development cost is k ; liquidation value is L .
- $t = 2$ (product-market competition): if development succeeds, the new product competes with incumbents under differentiated Bertrand competition.

2.2 Product-market profits and overlap

Notation uses subscript “acq” for acquisition and “:acq” for no acquisition. If the project succeeds (S) and is not acquired, the entrepreneur earns $\pi_{:acq,S}^E$ and the acquirer earns $\pi_{:acq,S}^A > 0$. If the project fails (F) and is not acquired, the entrepreneur earns $\pi_{:acq,F}^E = 0$ and the acquirer earns $\pi_{:acq,F}^A > 0$. If acquired and the project fails, the acquirer still sells its old product, so $\pi_{acq,F}^A = \pi_{:acq,F}^A$. If acquired and the project succeeds, the acquirer is a multiproduct price-setter with profit $\pi_{acq,S}^A$.

The paper summarizes the profit ranking as (their eq. (1)):

$$\pi_{acq,S}^A > \pi_{acq,F}^A = \pi_{:acq,F}^A > \pi_{:acq,S}^A > 0, \quad \pi_{:acq,S}^E > \pi_{:acq,F}^E = 0. \quad (1)$$

2.3 Development decision at $t = 1$ and the replacement effect

Define the incremental gain from successful development (relative to failure) for the entrepreneur and the acquirer:

$$D_E \equiv \pi_{acq,S}^E - \pi_{:acq,F}^E, \quad D_A \equiv \pi_{acq,S}^A - \pi_{acq,F}^A.$$

Under product-market overlap (imperfect substitutes), the incumbent's incremental gain is smaller because the new product cannibalizes the old one. The paper states (their eq. (2)):

$$D_E = \pi_{acq,S}^E - \pi_{acq,F}^E > \pi_{acq,S}^A - \pi_{acq,F}^A = D_A. \quad (2)$$

Interpretation: $D_E > D_A$ is the *replacement effect* (Arrow 1962): incumbents internalize cannibalization and therefore have weaker incentives to innovate.

2.4 Threshold form of development choices (eqs. (3)–(4))

Let $d_E, d_A \in \{0, 1\}$ denote development choices. The entrepreneur develops if expected net payoff exceeds liquidation:

$$r_E D_E - k \geq L, \quad \text{and the acquirer develops if} \quad r_A D_A - k \geq L. \quad (3)$$

Define development-cost thresholds:

$$k_E \equiv r_E D_E - L, \quad k_A \equiv r_A D_A - L. \quad (4)$$

Then E develops iff $k \leq k_E$ and A develops iff $k \leq k_A$.

2.5 Proposition 1: overlap reduces post-acquisition development

Proposition 1 (Project development and market overlap). *For any overlap, the entrepreneur continues development for $k \leq k_E$, while an incumbent acquirer continues for $k \leq k_A$. Holding development capabilities fixed, overlap reduces the acquirer's incentive to continue (via D_A), making k_A smaller than k_E unless synergies are sufficiently large.*

Empirical mapping: overlapping acquired projects should be *less likely* to exhibit development activity after acquisition.

2.6 Proposition 2: competition mitigates the replacement effect

Let n be the number of incumbent competitors. With more competitors, cannibalization losses from a new product are spread across firms, lowering the replacement effect.

Proposition 2 (Project development and competition). *For any overlap, the gap $k_E - k_A$ is strictly decreasing in n .*

Empirical mapping: the drop in development for overlapping acquisitions should be *stronger* in low-competition markets.

2.7 Proposition 3: longer patent life strengthens killer incentives

Let T_A denote remaining patent life of the incumbent's overlapping product. When T_A is large, the incumbent has more to lose from cannibalization; when T_A is near zero, generic entry erodes the old product's profits and weakens the replacement effect.

Proposition 3 (Project development and patent life). *For any overlap (and when development synergies are not too large), the gap $k_E - k_A$ is strictly increasing in T_A .*

Empirical mapping: overlapping acquisitions should be *more likely to be discontinued* when the acquirer’s relevant patent is far from expiration.

2.8 Acquisition decision at $t = 0$: efficiency effect vs replacement effect (eqs. (5)–(6))

Suppose the entrepreneur would develop if independent ($k \leq k_E$). The incumbent benefits from acquiring because it can prevent (or soften) future competition. This is the **efficiency effect**:

$$r_E(\pi_{acq,F}^A - \pi_{acq,S}^A).$$

But the incumbent must compensate the entrepreneur for her private value of continuing development; this is the **replacement effect** term. When $k_E \geq k > k_A$, the incumbent would *not* develop if it owns the project, so acquisition implies *acquire-to-kill*. The condition for acquire-to-kill is (their eq. (5)):

$$\underbrace{r_E(\pi_{acq,F}^A - \pi_{acq,S}^A)}_{\text{efficiency effect}} \geq \underbrace{(r_E D_E - k) - L}_{\text{replacement effect}}. \quad (5)$$

When $k \leq k_A$, both would develop, and acquisition is *acquire-to-continue*. The condition is (their eq. (6)):

$$\underbrace{r_E(\pi_{acq,F}^A - \pi_{acq,S}^A)}_{\text{efficiency effect}} \geq \underbrace{r_E D_E - r_A D_A}_{\text{replacement effect}}. \quad (6)$$

2.9 Proposition 4: overlap predicts acquisition

Proposition 4 (Acquisition). *The incumbent has strictly positive incentives to acquire when there is product-market overlap (or when development synergies are sufficiently large). Without overlap and without strong synergies, strategic acquisition motives are absent and acquisition is not profitable.*

Empirical mapping: overlap should *positively predict* acquisition.

2.10 Multiple incumbents (sketch)

With several potential acquirers, killing a future competitor resembles a costly private provision problem (volunteer’s dilemma). Acquisition becomes more likely when there are more potential acquirers (more potential bidders), and the winning acquirer tends to be the one with the strongest incentive (typically the greatest overlap and/or strongest synergies).

3 Data and measurement (how the paper maps theory to empirics)

3.1 Setting: pharmaceutical R&D

Drug development has clear milestones and is highly regulated. Observable “development events” are costly and informative about real investment. This makes project-year activity a credible proxy

for continuation vs termination.

3.2 Project data: Pharmaprojects

The paper uses Pharmaprojects (Pharma Intelligence) to follow drug projects from early stage through launch/discontinuation. Key descriptors include therapeutic class (TC), mechanism of action (MOA), originator firm, and time-varying development events.

3.3 Acquisition data and merging

Acquisitions are identified from multiple M&A datasets and merged to project data by firm matching. A project is treated as acquired if its originator firm is acquired during the project’s development life.

3.4 Overlap measure (core identification variable)

Overlap is defined at acquisition time as whether the acquirer already has a product/project in the same:

$$(\text{narrow}) \text{ overlap: } (\text{TC} \times \text{MOA}), \quad (\text{broader}) \text{ overlap: TC}.$$

Overlap operationalizes product-market substitutability and therefore the replacement effect.

3.5 Competition environment

Competition is measured as the number of firms with a drug project in the same market definition (typically $\text{TC} \times \text{MOA}$). The paper often splits the sample into high vs low competition at the median.

3.6 Patent life

For overlapping acquisitions, the paper matches the acquirer’s relevant patent and constructs an indicator

$$\mathbf{1}\{\text{NearPatExpiry}\} \equiv \mathbf{1}\{\text{within 5 years of patent expiration}\},$$

to test the prediction that shorter patent life weakens killer incentives.

3.7 Antitrust scrutiny: HSR threshold

US deals above the Hart–Scott–Rodino (HSR) size threshold are more likely to receive antitrust scrutiny. The paper tests whether suspicious bunching just below the threshold is associated with higher discontinuation.

4 Empirical strategy: all econometric models

4.1 Model (7): project-year development panel (main test)

Projects enter the panel in their origination year and exit at U.S. launch (if any). The main specification is a triple-difference panel regression (linear probability model; logit robustness noted in the paper):

$$\begin{aligned} Development_{i,t} = & \beta \mathbf{1}\{Acquired_i\} \mathbf{1}\{Post_{i,t}\} \mathbf{1}\{Overlap_i\} + \gamma_1 \mathbf{1}\{Acquired_i\} \mathbf{1}\{Post_{i,t}\} \\ & + \gamma_2 \mathbf{1}\{Acquired_i\} \mathbf{1}\{Overlap_i\} + \gamma_3 \mathbf{1}\{Acquired_i\} + \alpha_{FE} + \varepsilon_{i,t}. \end{aligned} \quad (7)$$

Coefficient of interest: β captures the *incremental* post-acquisition drop in development for *overlapping* acquisitions relative to nonoverlapping acquisitions and relative to the nonacquired baseline, conditional on fixed effects.

Fixed effects used across columns (Table 2): vintage FE, age FE, age $\times$ (TC $\times$ MOA) FE, originator (target) FE, and/or project FE. Standard errors are clustered at the project level. The paper also uses propensity-score reweighting as an additional selection-control device.

4.2 Competition heterogeneity (Table 4)

To test Proposition 2, the regression is interacted with a low-competition indicator:

$$\mathbf{1}\{Acquired\} \mathbf{1}\{Post\} \mathbf{1}\{Overlap\} \mathbf{1}\{LowComp\},$$

so that the overlap effect is allowed to differ between low- and high-competition markets.

4.3 Patent-life heterogeneity for overlapping acquisitions (Table 5)

Restricting to acquired overlapping projects, the paper estimates:

$$Development_{i,t} = \beta_0 \mathbf{1}\{Post_{i,t}\} + \beta_1 \mathbf{1}\{NearPatExpiry_i\} + \gamma \mathbf{1}\{Post_{i,t}\} \mathbf{1}\{NearPatExpiry_i\} + \alpha_{FE} + \varepsilon_{i,t}. \quad (1)$$

Prediction: $\gamma > 0$ (near-expiry weakens the incentive to kill, attenuating the post-acquisition drop).

4.4 Model (8): acquisition choice using conditional logit (Table 6)

To test Proposition 4, the paper compares realized acquirer–target pairs to pseudo-pairs. For each deal d , it forms sets of potential acquirers (random controls or size-matched controls) and estimates:

$$\mathbf{1}\{Acquirer-Target_{ij,d,t}\} = \beta \mathbf{1}\{Overlap_{ij,d,t}\} + \alpha_{DealFE} + \varepsilon_{ij,d,t}. \quad (8)$$

This is implemented as a conditional logit with deal fixed effects, and the paper reports marginal effects. An interaction with low competition tests whether overlap-driven acquisition is stronger when market power is larger.

4.5 Antitrust threshold discontinuity evidence (Table 7 and Figure 3)

This evidence is primarily univariate: compare post-acquisition status (active/launched/discontinued) for deals just below vs just above the HSR review threshold, and use a placebo pseudothreshold to rule out smooth size-value relationships.

4.6 Asset redeployment and human-capital channel (Table 8)

Panel A uses a pair dataset (acquired drug $\times$ acquirer-originated drug within a 5-year window) and estimates:

$$Outcome = \beta \mathbf{1}\{Post\} \mathbf{1}\{Overlap\} + \gamma \mathbf{1}\{Post\} + \delta \mathbf{1}\{Overlap\} + FE + \varepsilon,$$

where outcomes are chemical similarity (Tanimoto) and citations to targets. Panel B uses t-tests comparing inventor patenting before/after acquisition for inventors who join the acquirer vs those who move elsewhere.

4.7 Acquisition value regression (Table 9)

The paper estimates:

$$\ln(Value_i) = \beta \mathbf{1}\{Overlap_i\} + \alpha_{FE} + \varepsilon_i,$$

to test whether overlapping acquisitions are purchased at lower prices.

5 Results

5.1 Figure 1: model time line

Message: the model separates (i) acquisition choice, (ii) continuation vs termination decision, and (iii) product-market competition. This separation maps cleanly to the empirical strategy: overlap affects incentives at the *continuation* stage and also predicts acquisition.

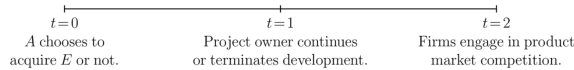


FIG. 1.—Model time line.

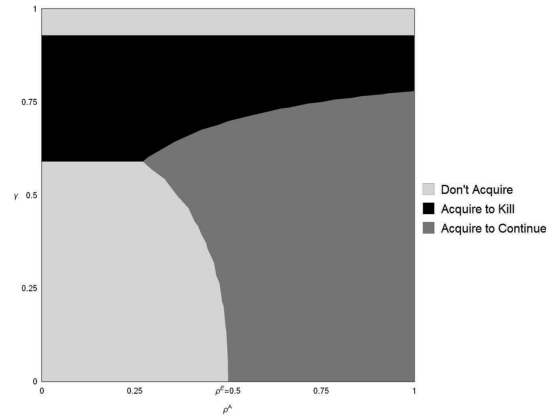


FIG. 2.—Optimal acquisition strategies. This graph plots the optimal acquisition decisions—“Don’t Acquire” (light gray), “Acquire to Kill” (black), and “Acquire to Continue” (dark gray)—as functions of the acquirer’s development capability ρ^A and the degree of substitutability γ . The other parameter values are $\alpha^A = \alpha^B = 100$, $\rho^B = 0.5$, $L = 20$, $k = 80$, and $n = 2$.

5.2 Figure 2: optimal acquisition strategies

Message: the acquire/kill/continue regions depend on two forces:

- stronger overlap (higher substitutability) increases the replacement effect and therefore expands the “acquire-to-kill” region;
- stronger development capability r_A expands the “acquire-to-continue” region (synergies dominate).

Economic lesson: killing can be optimal even when the incumbent is good at development, because overlap makes continued development privately unattractive.

5.3 Table 1: descriptive acquisition rates

What it is: counts of drug projects and shares that are nonacquired, acquired by nonoverlapping acquirers, and acquired by overlapping acquirers. **How to read it:**

- Overall, most projects are not acquired; a meaningful minority are acquired; and a smaller subset are acquired by overlapping incumbents.
- Acquisition rates are lower for more recent originations, consistent with right-truncation (projects are younger and have had less time to be acquired).

Why it matters: it establishes that overlapping acquisitions exist in the data and are economically nontrivial.

	N	Nonacquired (%)	Nonoverlapping Acquired (%)	Overlapping Acquired (%)
Whole sample	16,015	78	17	5
By time period:				
Beginning–1995	2,684	60	31	9
1996–2000	2,854	68	25	7
2001–5	4,716	79	16	4
2006–10	5,761	90	8	2
By high-level disease group (top 5):				
Anticancer (13 TCs; 783 TC-MOAs)	2,579	80	16	4
Neurological (27 TCs; 986 TC-MOAs)	2,573	77	19	4
Anti-infectives (28 TCs; 452 TC-MOAs)	1,946	77	16	7
Biotechnology (26 TCs; 209 TC-MOAs)	1,493	79	16	5
Alimentary/metabolism (24 TCs; 498 TC-MOAs)	1,380	81	15	4

	DEVELOPMENT EVENT = 1					
	(1)	(2)	(3)	(4)	(5)	(6)
$I(\text{Acquired}) \times I(\text{Post}) \times \text{Overlap}$	-.037*** (.013)	-.033** (.014)	-.029* (.015)	-.041** (.019)	-.043** (.021)	-.054** (.024)
$I(\text{Acquired}) \times I(\text{Post})$	-.020*** (.006)	-.016** (.007)	-.017** (.009)	-.024** (.010)	-.018 (.011)	-.018 (.013)
$I(\text{Acquired}) \times \text{Overlap}$	.004 (.008)	.009 (.009)	.026** (.011)			
$I(\text{Acquired})$	-.002 (.004)	-.004 (.005)	-.011 (.012)			
Before(-3) $\times$ Overlap						-.031 (.032)
Before(-2) $\times$ Overlap						.012 (.032)
Before(-1) $\times$ Overlap						-.040 (.030)
Before(-3)						.015 (.017)
Before(-2)						.020 (.017)
Before(-1)						-.003 (.016)
Observations	143,569	143,569	143,569	143,569	134,662	143,569
R^2	.038	.252	.289	.366	.662	.370
Vintage FE	Y	Y	Y			
Age FE	Y					
Age $\times$ TC $\times$ MOA FE		Y	Y	Y	Y	Y
Originator (target company) FE			Y			
Project FE				Y	Y	Y
Propensity score reweighted					Y	

5.4 Table 2: main development result (equation (7))

Design: project-year panel; dependent variable is a dummy for any development event in a year.

Key coefficient: β on $1\{\text{Acquired}\}1\{\text{Post}\}1\{\text{Overlap}\}$ is **negative and statistically significant**

across specifications. **Interpretation:** after acquisition, overlapping projects are substantially *less likely* to have development activity than nonoverlapping acquired projects.

Pre-trends check (leads). Table 2 includes “Before($-t$) $\times$ Overlap” terms. The lack of strong negative pre-trends supports a parallel-trends style interpretation: overlapping and nonoverlapping targets do not show systematically lower activity *before* acquisition, but diverge *after* acquisition.

Fixed effects escalation. As the specification becomes more demanding (originator FE, project FE, age $\times$ TC $\times$ MOA FE), the overlap effect remains negative, suggesting it is not driven by time-invariant target quality or by market-time composition.

5.5 Table 3: robustness and “never developed” evidence

Table 3 strengthens the main interpretation in three ways:

1. **Acquired-only sample:** within acquired projects, overlap still predicts a lower likelihood of post-acquisition development events.
2. **Single-project targets:** the overlap effect is even larger when the target has only one project at acquisition time, ruling out the simple “acquirer reprioritizes within a multi-project target” story.
3. **Never-developed outcome:** overlapping acquisitions are more likely to have *no post-acquisition development at all*, consistent with true termination rather than mere delay or reshuffling of milestones.

TABLE 3
OVERLAPPING ACQUISITIONS AND PROJECT DEVELOPMENT: ALTERNATIVE SPECIFICATIONS

	DEVELOPMENT EVENT = 1			No DEVELOPMENT = 1
	(1)	(2)	(3)	(4)
$I(\text{Acquired}) \times I(\text{Post}) \times \text{Overlap}$	-.050** (.023)	-.121*** (.060)	.005 (.035)	.149*** (.033)
$I(\text{Acquired}) \times I(\text{Post})$	-.024 (.015)	-.041 (.025)	-.095*** (.013)	.401*** (.021)
Observations	27,784	19,651	7,916	9,227
R^2	.445	.249	.155	.477
Sample	Acquired projects only	Single-project target only	Excluding “never developed”	
TC $\times$ MOA FE				Y
Age $\times$ TC $\times$ MOA FE	Y	Y	Y	
Project FE	Y	Y	Y	Y

TABLE 4
OVERLAPPING ACQUISITIONS AND PROJECT DEVELOPMENT: MARKET COMPETITION

	DEVELOPMENT EVENT = 1			No DEVELOPMENT = 1		
	Low (1)	High (2)	Interacted (3)	Low (4)	High (5)	Interacted (6)
$I(\text{Acquired}) \times I(\text{Post}) \times \text{Overlap}$	-.065** (.026)	.017 (.035)	.017 (.035)	.219*** (.054)	.038 (.070)	.038 (.070)
$I(\text{Acquired}) \times I(\text{Post}) \times \text{Overlap} \times \text{LowCompetition}$			-.082* (.044)			.181** (.089)
Observations	74,261	69,308	143,569	5,991	3,236	9,227
R^2	.415	.399	.408	.497	.474	.489
TC $\times$ MOA FE				Y	Y	Y
Age $\times$ TC $\times$ MOA FE	Y	Y	Y	Y	Y	Y
Project FE	Y	Y	Y	Y	Y	Y

5.6 Table 4: competition heterogeneity (Proposition 2)

Finding: the post-acquisition development decline for overlapping projects is concentrated in **low-competition** markets. In high-competition markets the overlap coefficient is small and not significant, while in low-competition markets it is large and negative. **Interpretation:** when incumbents have more market power, the incentive to preempt competition is stronger.

5.7 Table 5: patent-life heterogeneity (Proposition 3)

Sample: acquired *overlapping* projects only. **Key interaction:** $1\{Post\} \times 1\{NearPatExpiry\}$ is **positive and significant**. **Interpretation:** when the incumbent’s relevant patent is close to expiration, cannibalization concerns are weaker, so the post-acquisition suppression of development is attenuated (and can even reverse). This matches the theory: long remaining patent life strengthens the replacement effect and makes killing more attractive.

TABLE 5
OVERLAPPING ACQUISITIONS AND PROJECT DEVELOPMENT:
ACQUIRER’S PATENT LIFE

	DEVELOPMENT EVENT = 1	
	(1)	(2)
$I(Post) \times I(NearPatExpiry)$	.013 (.133)	.406*** (.090)
$I(Post)$	-.173* (.092)	-.210*** (.067)
$I(NearPatExpiry)$	-.104** (.043)	-.147*** (.043)
Observations	6,398	6,398
R^2	.212	.450
Vintage FE	Y	Y
Age FE	Y	Y
TC $\times$ MOA FE	Y	
Age $\times$ TC $\times$ MOA FE		Y

TABLE 6
PRODUCT OVERLAP AND ACQUISITION DECISIONS

	ACQUISITION = 1				ACQUISITION = 1			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Overlap (TCMOA)	.629*** (.009)		.577*** (.015)		.194*** (.010)		.209*** (.015)	
Overlap (TC)		.356*** (.005)		.300*** (.008)		.214*** (.008)		.200*** (.011)
Overlap (TCMOA) $\times$ LowCompetition			.088*** (.019)				-.027 (.020)	
Overlap (TC) $\times$ LowCompetition				.103*** (.011)				.025* (.013)
Observations	55,374	55,374	38,430	38,430	34,005	34,005	34,005	34,005
Pseudo R^2	.118	.119	.098	.097	.032	.064	.052	.065
Deal FE	Y	Y	Y	Y	Y	Y	Y	Y
Matching method			Random matching				Matched by pipeline size	
No. of deals	9,229	9,229	9,229	9,229	9,229	9,229	9,229	9,229
No. of control deals	46,145	46,145	46,145	46,145	46,145	46,145	46,145	46,145

5.8 Table 6: overlap predicts acquisition (Proposition 4)

Design: conditional logit comparing realized acquirer–target matches to pseudo-deals, with deal fixed effects. **Finding:** overlap (both narrow TC–MOA and broader TC) strongly increases the likelihood that a firm is the acquirer. **Interpretation:** acquisitions are disproportionately directed at targets that threaten the acquirer’s product space, consistent with strategic acquisition motives (though not alone sufficient to prove killing).

5.9 Figure 3 and Table 7: antitrust threshold evidence

Figure 3: shows bunching of acquisition deal sizes just below the HSR review threshold, consistent with avoiding scrutiny. **Table 7:** compares outcomes for deals just below vs above the threshold. Deals just below are **much more likely to be discontinued** and less likely to be launched. A placebo “pseudothreshold” shows no comparable discontinuity. **Interpretation:** this pattern is hard to reconcile with smooth deal-size/quality relationships alone and is consistent with strategic avoidance of antitrust review for killer-type deals.

5.10 Table 8: post-acquisition asset redeployment and human capital

Panel A (similarity and citations). The interaction $1\{Post\} \times 1\{Overlap\}$ is small and not significant for chemical similarity and citation outcomes. **Interpretation:** there is little evidence that acquirers systematically redeploy the killed project’s assets into closely related internal projects. This weakens a “benign reprioritization” story where resources shift to a better overlapping internal line.

Panel B (inventor mobility and productivity). Inventors who move to the acquirer show a large drop in patenting after acquisition relative to before, while inventors who move elsewhere show

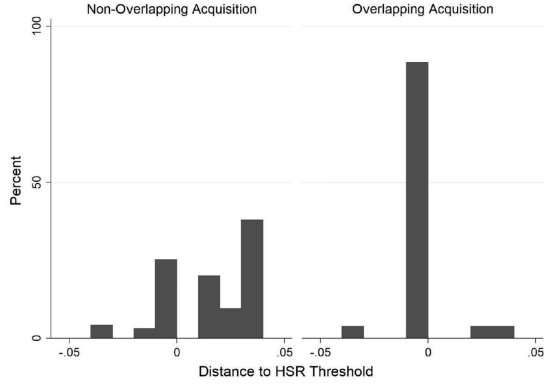


FIG. 3.—Distribution of acquisition size near the Hart-Scott-Rodino (HSR) review threshold. Acquisitions that fall into the interval $[-5\%, 5\%]$ around the threshold are kept, and the horizontal axis represents the distance to the review threshold (from -5% to 5%). Nonoverlapping acquisitions are reported on the left, and overlapping acquisitions are reported on the right.

TABLE 7
INTENSITY OF PROJECT DISCONTINUATION AROUND FTC REVIEW THRESHOLD

	5% below Threshold (%)	5% above Threshold (%)	Difference (%)	T-Statistics	Statistical Significance
A. Real HSR Threshold					
Active	3.57	7.58	-4.00	-1.18	Not significant
Launched	1.79	9.09	-7.31	-2.29	5% level
Discontinued	94.64	83.33	11.31	2.51	5% level
B. Pseudothreshold					
Active	7.41	2.63	4.78	1.20	Not significant
Launched	3.70	4.39	-.69	-.16	Not significant
Discontinued	88.88	92.98	-4.10	-.71	Not significant

a smaller decline. **Interpretation:** acquisitions do not simply absorb and intensify innovative effort; human capital appears to be underutilized at the acquirer, consistent with shelving incentives.

TABLE 8
POSTACQUISITION ASSET REDEPLOYMENT

	(1)	(2)	(3)	(4)	(5)	(6)
A. Project Similarities to Acquired Drugs before and after Acquisition						
	Chemical Similarity			Clonidine to Targets		
$I(\text{Post}) \times \text{Overlap}$	.001 (.003)	.000 (.003)	.002 (.002)	-.002 (.002)	-.002 (.002)	-.000 (.000)
$I(\text{Post})$	-.002 (.004)	-.001 (.004)	-.004 (.003)	.000 (.000)	.001 (.001)	.000 (.000)
Overlap	.004 (.003)	.004 (.003)	.002 (.002)	.002 (.002)	.002 (.002)	.000 (.000)
Observations	154,896	154,896	154,896	154,896	154,896	154,896
R^2	.001	.014	.263	.001	.004	.154
Acquirer FE		Y			Y	
Case FE			Y			Y
B. Inventor Mobility and Patent Productivity before and after Acquisition						
	Before Acquisition		After Acquisition		Difference	
Inventors who move to acquirer after acquisition (22%)	4.572		3.160		-1.412***	
Inventors who move to other firms after acquisition (78%)	4.337		4.089		-.207*	
Difference	-.215		.920***		1.144***	

TABLE 9
PROJECT OVERLAP AND ACQUISITION VALUE

	ln(Acquisition Value)		
	(1)	(2)	(3)
Overlap	.126 (.101)	.025 (.067)	-.082 (.114)
Observations	14,660	14,660	14,660
R^2	.844	.905	.940
Acquirer FE	Y	Y	Y
Age FE	Y	Y	
TC $\times$ MOA FE		Y	
Age $\times$ TC $\times$ MOA FE			Y

5.11 Table 9: overlap and acquisition value

Finding: overlap is not robustly associated with lower acquisition value once rich fixed effects are included. **Interpretation:** the results are not driven by overlapping targets being systematically “cheaper lemons”; overlap does not simply proxy for low target quality priced into the deal.

5.12 Back-of-the-envelope prevalence of killer acquisitions (from text around Table 9)

The paper provides a conservative mixture calculation. Let the adjusted continuation rate for overlapping acquired projects be $p^{over} = 13.4\%$ and for nonoverlapping acquired projects be $p^{non} = 17.5\%$. Assume overlapping acquisitions are a mixture:

$$p^{over} = \nu \cdot 0 + (1 - \nu) \cdot p^{non},$$

so

$$\nu = 1 - \frac{p^{over}}{p^{non}}.$$

This yields $\nu \approx 23.4\%$ of overlapping acquisitions being “pure killers.” Multiplying by the share of acquisitions that are overlapping (about 22.7% in their calculation) implies about 5.3% of all

acquisitions are killers. Using a higher benchmark for nonkiller continuation (e.g., nonacquired projects) raises this estimate to about 7.4%.

6 What to remember (exam-style takeaways)

1. **Mechanism:** overlap $\Rightarrow$ replacement effect $\Rightarrow$ weaker post-acquisition continuation incentives.
2. **Heterogeneity:** killer incentives are strongest where incumbents have market power (low competition) and where cannibalization is valuable (long patent life).
3. **Identification structure:** project-year panel with rich FE + pre-trend checks + acquisition-choice conditional logit + antitrust-threshold evidence.
4. **Policy relevance:** the deals most likely to be killers appear to cluster just below antitrust scrutiny thresholds.

7 Conclusion

7.1 Summary

The paper concludes that **some acquisitions of innovative targets are motivated by the desire to eliminate future competition and are followed by the discontinuation of the target's innovation project.** These transactions—*killer acquisitions*—are most likely when the acquired innovation **overlaps** with the acquirer's product portfolio, when product-market **competition is weak** (high incumbent market power), and when the incumbent's relevant **patent protection is far from expiration.**

The argument is built as a unified chain:

1. **Theory:** overlap creates a *replacement effect*. An incumbent internalizes cannibalization of its existing product, so conditional on owning the project, it has weaker incentives to continue development than an independent entrepreneur.
2. **Empirics:** in pharmaceutical project-level data, acquired projects are significantly less likely to show post-acquisition development activity when they overlap with the acquirer's portfolio, with no comparable divergence *before* acquisition.
3. **Strategic selection:** overlap also predicts *which* acquirer wins the deal (conditional-logit evidence), consistent with strategic motives rather than purely synergistic matching.
4. **Policy relevance:** suspicious discontinuation is disproportionately concentrated among deals just below antitrust review thresholds, consistent with avoidance of scrutiny.

7.2 Key empirical conclusions

The paper's main empirical conclusions can be stated compactly:

- **Post-acquisition discontinuation is higher with overlap.** In project-year panels with rich fixed effects, the incremental post-acquisition drop in development activity is significantly larger for overlapping acquisitions than for nonoverlapping acquisitions.
- **The overlap effect is strongest where incumbents have more market power.** The post-acquisition reduction in development for overlapping acquisitions is concentrated in *low-competition* markets, consistent with a stronger incentive to preempt entry.
- **Patent life matters in the predicted direction.** For overlapping acquisitions, discontinuation is attenuated when the acquirer’s relevant patent is near expiration, consistent with weaker cannibalization incentives as the incumbent’s protected rents fade.
- **Overlap predicts acquisition.** In deal fixed-effect (conditional-logit) designs comparing realized acquirer–target matches to plausible alternatives, overlap strongly increases the likelihood of being the acquirer.
- **Antitrust thresholds appear to matter.** There is bunching of deals just below the HSR threshold and discontinuation outcomes are worse just below the threshold than just above, consistent with strategic avoidance of review for the most problematic deals.

7.3 Quantitative takeaway: how prevalent are killer acquisitions?

The paper provides conservative back-of-the-envelope estimates based on adjusted continuation rates. Let overlapping acquired projects continue at an adjusted rate of **13.4%**, while nonoverlapping acquired projects continue at **17.5%**. If overlapping acquisitions are a mixture of (i) *pure killers* with 0% continuation and (ii) *nonkillers* with continuation equal to 17.5%, then the implied share of pure killers among overlapping acquisitions is **23.4%**. Multiplying by the share of acquisitions that overlap yields an estimate of roughly **5.3%** of all acquisitions being killer acquisitions. Under an alternative benchmark for nonkiller continuation (using nonacquired projects), the estimate rises to about **7.4%**. A separate lower-bound calculation using a “never developed after acquisition” gap implies at least **29** killer acquisitions per year in their scaling.

7.4 Why overlap creates “acquire-to-kill” incentives

The intuition is the replacement effect translated into an acquisition setting:

1. **Without ownership**, the entrant’s innovation threatens to reduce incumbent profits by increasing competition.
2. **With ownership**, successful development of the target’s product can cannibalize the incumbent’s own product profits. The incumbent therefore values success less than the entrepreneur does, especially when products overlap.
3. **Acquisition changes the objective function:** instead of maximizing the stand-alone value of the new project, the incumbent maximizes the value of its *whole portfolio*. If the competitive harm from entry is large and the cannibalization loss from success is large, then the incumbent prefers to buy the project and discontinue it.

4. **Market power amplifies the incentive:** when competition is weak, the incumbent’s rents are large, so preventing future competition is more valuable.
5. **Long patent life amplifies the incentive:** when protected rents are expected to last longer, cannibalization concerns persist, strengthening the motive to kill.

7.5 Why this matters for antitrust and IO inference

The paper implies that standard antitrust screening that focuses on *current* product-market overlap and on deal size can miss important harm:

- harm may operate through **loss of potential competition** (future products not yet on the market),
- harmful deals may be **small** (early-stage targets) and thus fall below thresholds,
- observed innovation outcomes (continuation/termination) provide a measurable, economically relevant endpoint for enforcement.

7.6 Takeaway

In pharmaceuticals, acquisitions of overlapping innovative targets systematically reduce post-acquisition development activity—especially in low-competition markets and when incumbents have long remaining patent life—and overlap predicts acquisition and bunching below antitrust thresholds, implying a nontrivial share of deals are best interpreted as “killer acquisitions” that remove future competition by terminating innovation.